



SURDS: Self-Supervised Attention-guided Reconstruction and Dual Triplet Loss for Writer Independent Offline Signature Verification

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Overview:

- In this work, we propose a two-stage deep learning framework that leverages self-supervised representation learning as well as metric learning for writer-independent offline signature verification.
- First, we train an image reconstruction network augmented by a patch-based 2D spatial attention mechanism.
- Next, the pre-trained encoder is fine-tuned with a projector head using a metric learning framework, whose objective is to optimize a dual triplet loss by sampling negative samples from both within the same writer class as well as from other writers.
- The proposed two-stage framework has been evaluated on two publicly available offline signature datasets and compared with various SOTA methods, where it has yielded promising results outperforming several existing pieces of work.

Contributions:

- A self-supervised attention-guided reconstruction-based pre-training strategy has been proposed for learning fine-grained representations from signature images.
- A dual triplet loss-based fine-tuning is also proposed to tackle both intra-writer and cross-writer negative samples.
- To the best of our knowledge, this is the first work to employ self-supervised learning concept in the context of OSV.

Proposed Model:

Preprocessing:

- We compute center of mass of the original image and then find the tightest bound of the foreground signature pixels, followed by cropping and resize to 256x256 pixels.

Self-supervised pre-training:

- The reconstruction is augmented by a patch-wise attention module of signature patches with respect to the global image representation.
- The patch-wise attention module enforces focusing on visually important parts of the signature in a local region-wise fashion.
- The pre-training objective is a simple MSE reconstruction loss, given by:

$$\mathcal{L}_{SSL} = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W |I_{ij} - Y_{ij}|^2$$

Dual triplet loss based fine-tuning:

- The pre-trained encoder is then fine-tuned using a dual triplet loss objective that leverages both intra-writer and cross-writer negative samples.

$$\begin{aligned} \mathcal{L}_{IW} &= \max\{0, \delta(z_A, z_P) - \delta(z_A, z_{N_{IW}}) + \mu\} \\ \mathcal{L}_{CW} &= \max\{0, \delta(z_A, z_P) - \delta(z_A, z_{N_{CW}}) + \mu\} \\ \delta(z_i, z_j) &= \|z_i - z_j\|_2 \quad \mathcal{L}_{DT} = \mathcal{L}_{IW} + \mathcal{L}_{CW} \end{aligned}$$

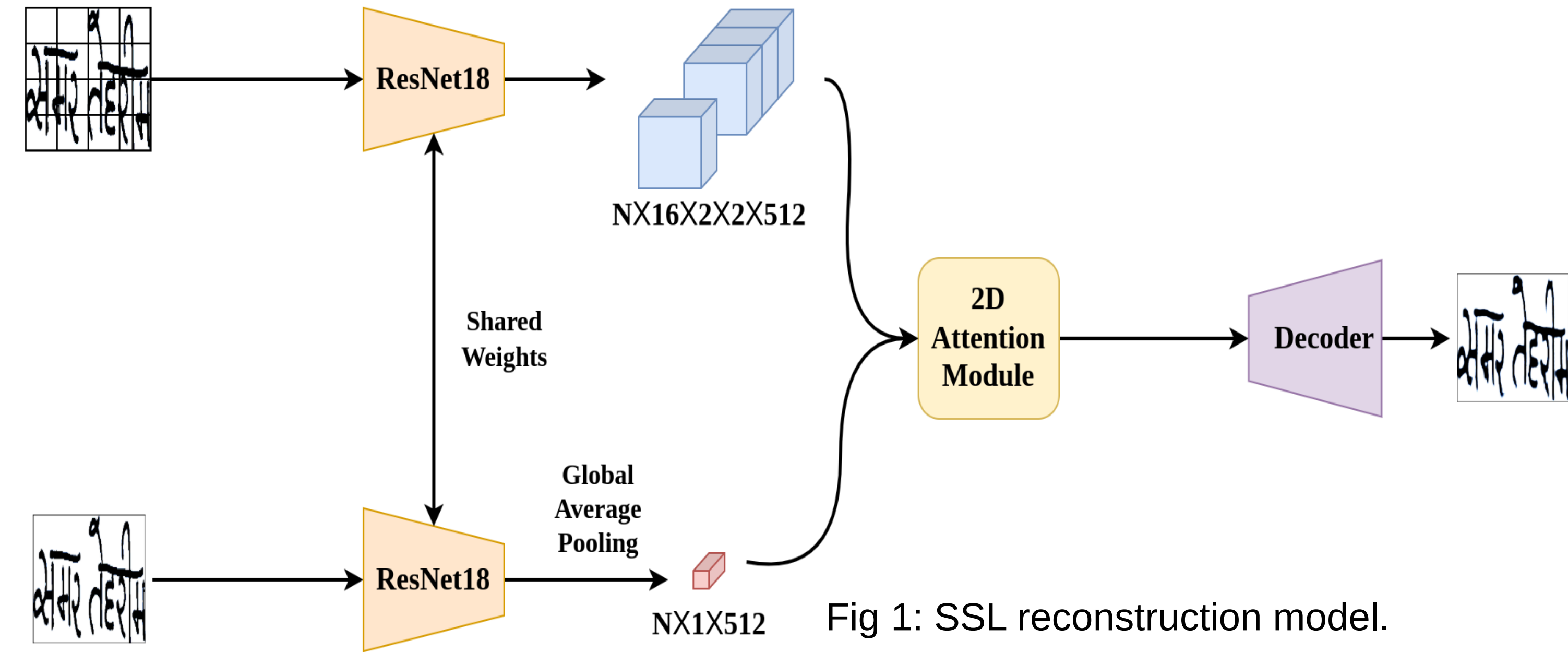


Fig 1: SSL reconstruction model.

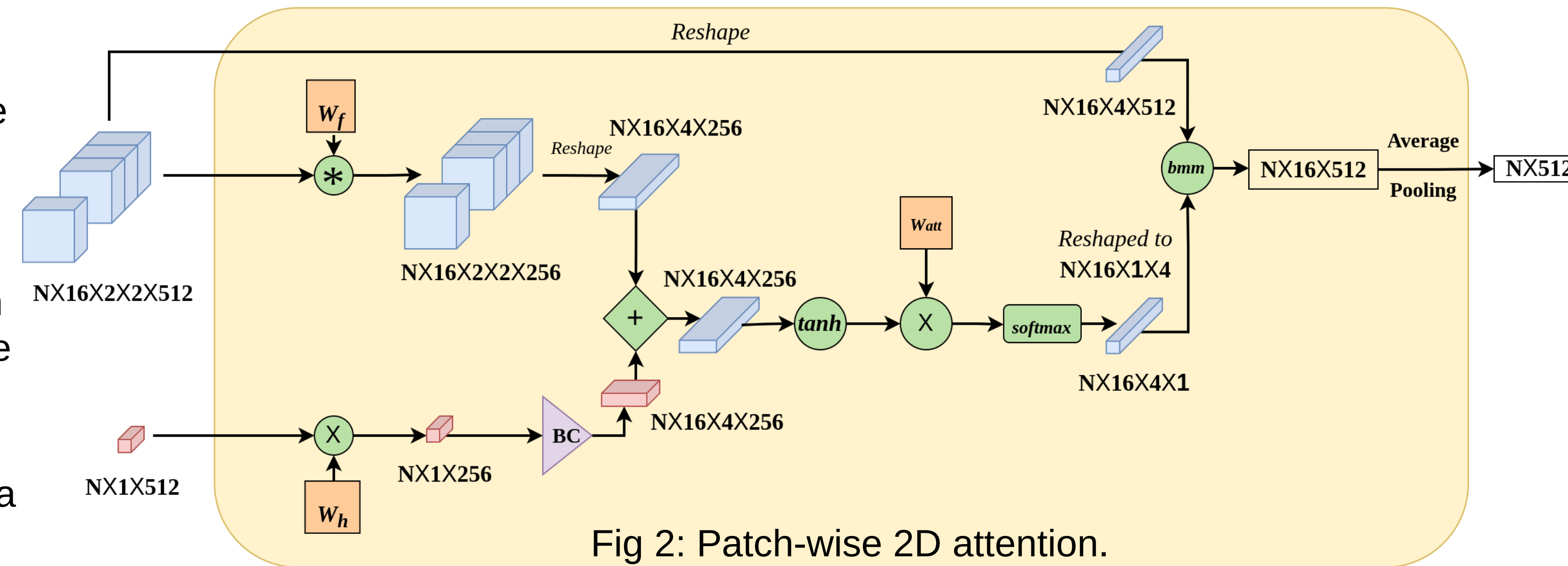


Fig 2: Patch-wise 2D attention.

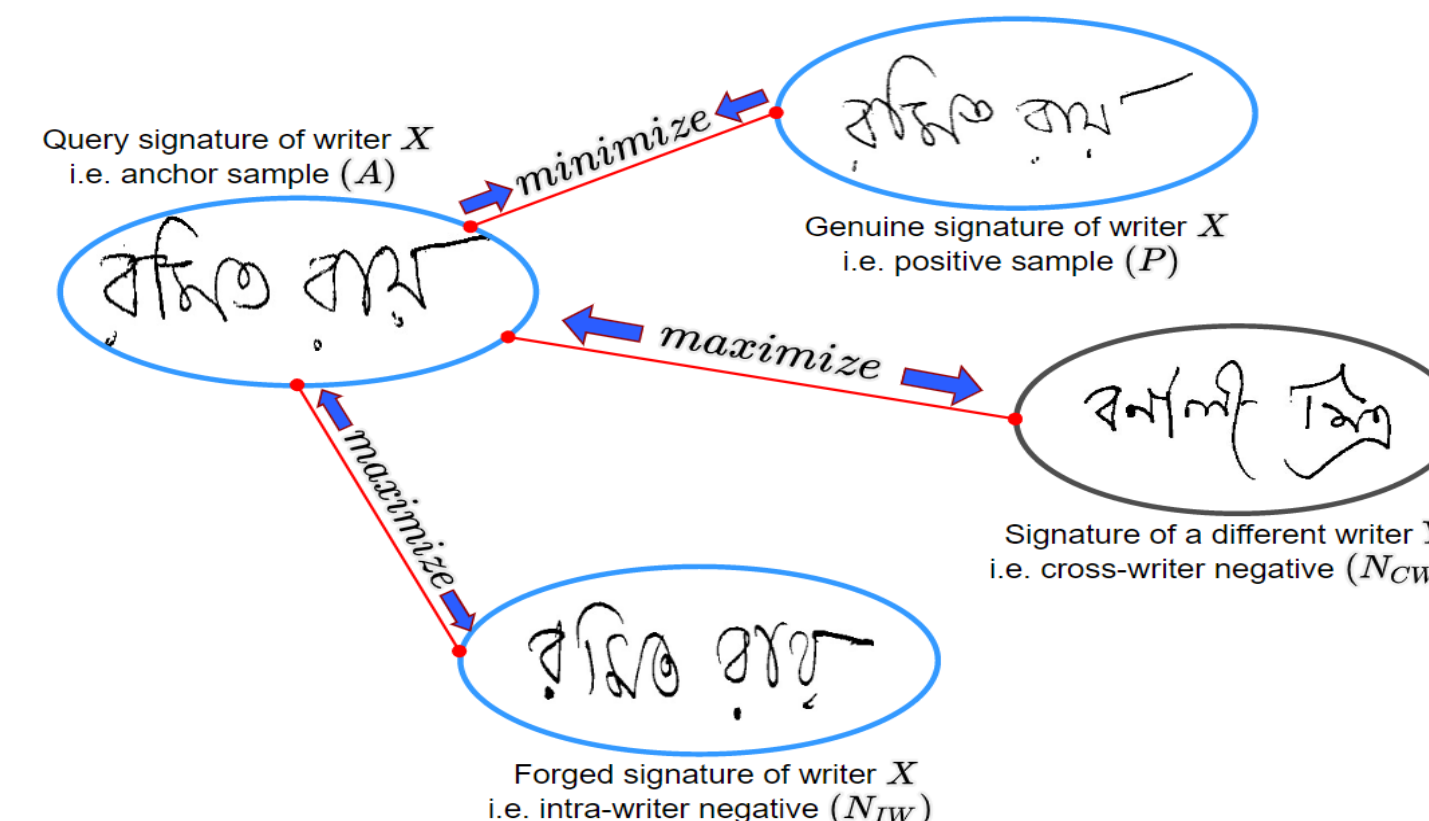
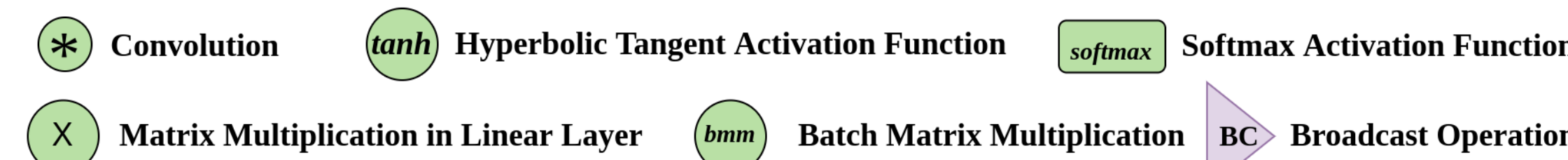


Fig 3: Dual triplet loss intuition.

Results and Analysis:

- Datasets: BHSig260 Hindi and Bengali, ICDAR-2011 Dutch signatures
- Evaluated against (1) SOTA SSL algorithms and (2) SOTA supervised methods. Refer to paper for more details.

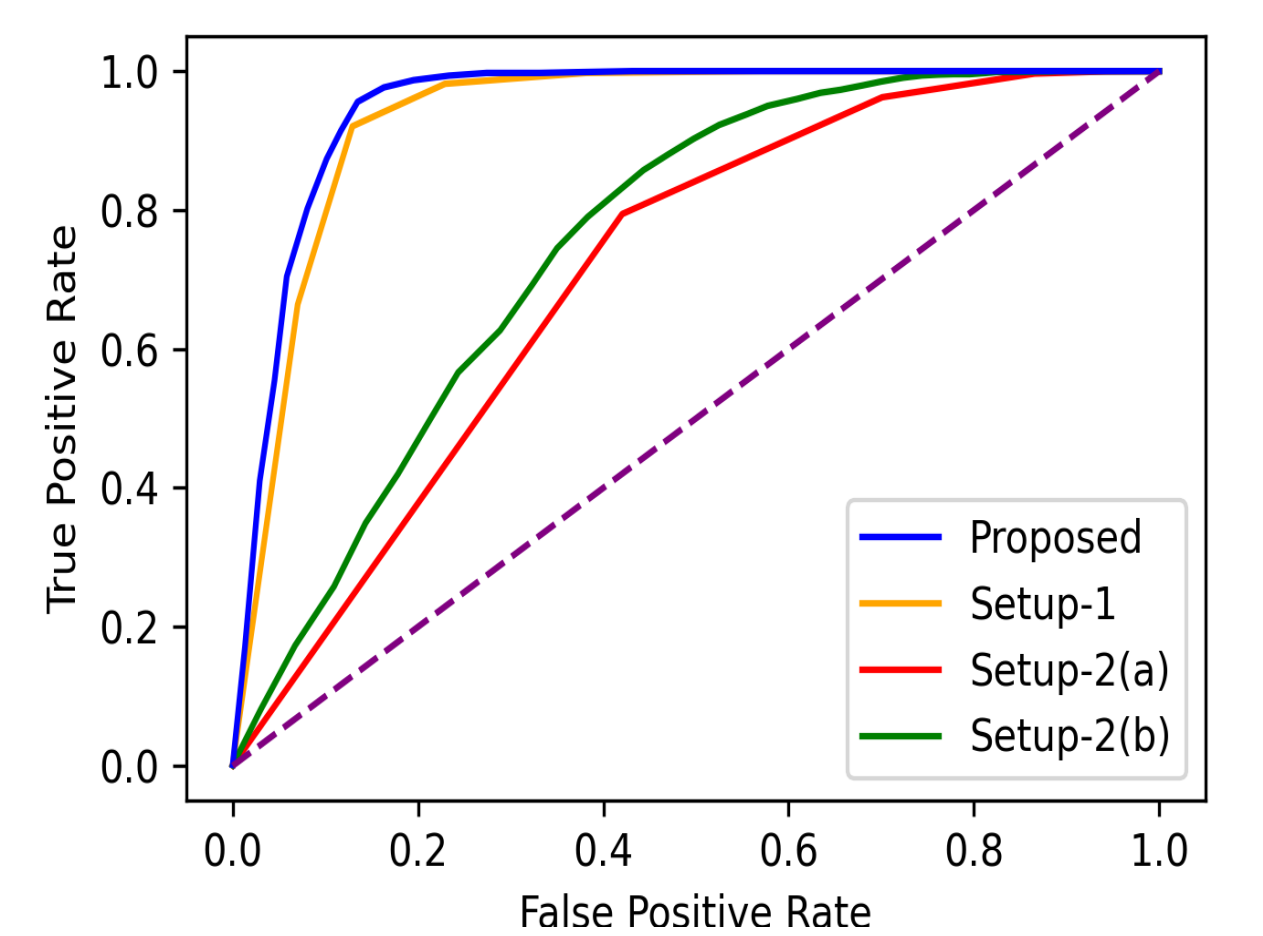
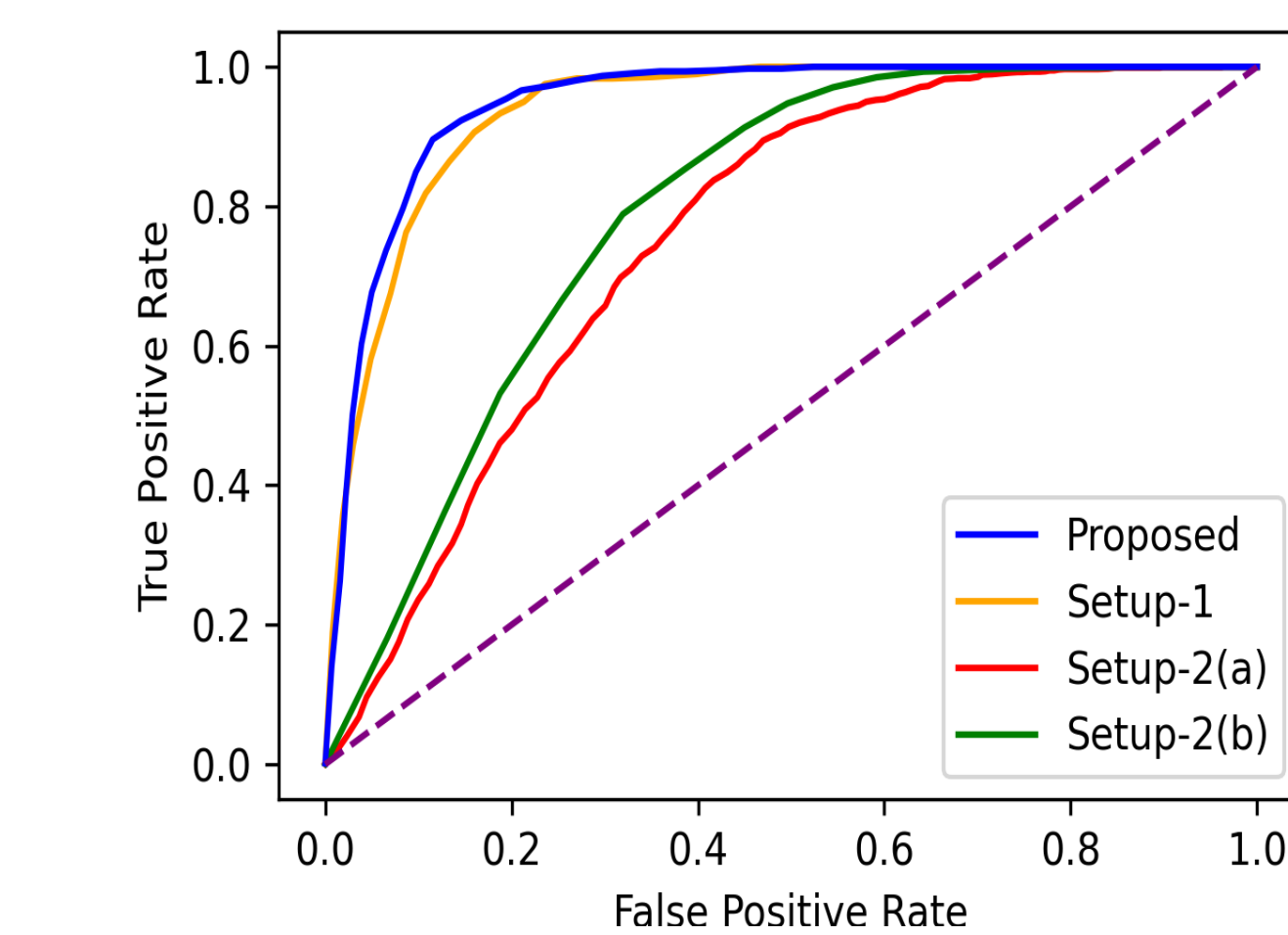
Method	BHSig260 Bengali [9]			BHSig260 Hindi [9]		
	Accuracy (%)	FAR (%)	FRR (%)	Accuracy (%)	FAR (%)	FRR (%)
SimCLR [6]	81.86	30.22	6.04	66.49	39.79	27.21
Barlow Twins [24]	82.75	28.44	6.25	72.20	42.53	13.06
SimSiam [23]	82.61	27.88	6.87	73.05	29.16	24.73
Pal et al. [9]	66.18	33.82	33.82	75.53	24.47	24.47
Dey et al. [8]	86.11	13.89	13.89	84.64	15.36	15.36
Dutta et al. [12]	84.90	15.78	14.43	85.90	13.10	15.09
Alaei et al. [13]*	—	16.18	30.12	—	16.18	30.12
Rateria et al. [18]	75.06	27.81	21.74	89.33	10.93	10.02
Jadhav et al. [32]*	67.00	—	—	67.00	—	—
Bhunia et al. [33]*	—	24.10	26.00	—	24.10	26.00
Jain et al. [34]	76.03	—	—	83.50	—	—
Proposed Method	87.34	19.89	5.42	89.50	12.01	8.98

Ablation study.

Baseline	BHSig260 Bengali [9]			BHSig260 Hindi [9]		
	Accuracy (%)	FAR (%)	FRR (%)	Accuracy (%)	FAR (%)	FRR (%)
RN-DTL	82.18	21.66	13.95	86.72	14.67	11.88
AE-DTL	83.57	29.11	3.75	89.14	18.47	3.25
Proposed Method	87.34	19.89	5.42	89.50	12.01	8.98

Cross-dataset train/test scenarios.

Train/Test	Bengali/Hindi			Hindi/Bengali		
	Accuracy (%)	FAR (%)	FRR (%)	Accuracy (%)	FAR (%)	FRR (%)
Setup-1	88.19	20.62	2.99	86.14	17.11	10.01
Setup-2(a)	67.01	51.14	14.82	72.26	46.46	9.00
Setup-2(b)	69.59	41.87	14.65	73.58	30.03	22.81



ROC curves for different setups on Bengali (left) and Hindi (right) datasets.